

Cover Letter

Dear Teacher,

I, Dr. Muhammad Bilal have currently completed Ph.D. (Information and Communication Engineering) from Harbin Engineering University. I have received B.S. (Telecommunication Engineering) and M.E. (Telecommunication Engineering) from FAST-NU and Hamdard University Pakistan in 2009 and 2013 respectively. In 2009, I started working in one of the largest telecom organization of Pakistan, PTCL, Etisalat as Assistant Manager (Access Networks). In 2011, I joined NESCOM, the largest research and development institute of Pakistan. We research and developed tremendous technical products for Pakistan Navy and other international Navys. I have developed and delivered various projects which includes underwater imaging and illumination system, underwater vehicles, tactical communication networks, underwater communication through software define radios and MODEMs.

In 2016 I started pursuing Ph.D. in College of Underwater Acoustic Engineering from Harbin Engineering University (HEU), Harbin, China. My research interests' lies in the areas of covert biologically inspired underwater acoustic communication and networking. My research is based on mimicry covert underwater acoustic communication mimicking sea natural noise. It is a newly flanged thought of low probability of recognition constraint covert communication. The waveform is made akin to natural sounds present in the environment making a perfect secure communication. I have published 15 papers in SCI and EI journals and conferences. List of published papers can be witnessed from my attached CV or Google scholar profile. I have been an active reviewer of various journals which includes IEEE Access, Journal of Ocean Engineering, Nature Scientific reports, Sensors, International Journal of Communication Systems, Mobile Information Systems, Aqua Culture and Fisheries and other journals. I have also got best cadre student award in Harbin Engineering University in year 2017-2018.

Research gate Profile: https://www.researchgate.net/profile/Muhammad_Bilal109/publications

Google Scholar Profile: <https://scholar.google.com/citations?user=51HfnB8AAAAJ&hl=en>

I wish to work under your valuable guidance as Assistant Professor/Post-Doctoral Researcher or any suitable position in your esteem lab. I have read your lab research on underwater acoustic communication and found very interesting. I wish to work on machine learning based cyber security techniques using mimicry covert communication. It is a novel technique and has a lot of research potential to be addressed. My research focuses on mimicry communication where an eavesdropper is unable to recognize the communication signal as a communication waveform. The enemy remains unaware of communication and assumes it as natural noise present in the environment. This research is of great importance for UAVs, UUVs, underwater nodes, and

IOTs, IOUTs, WSN and all types of sensor networks. It would be great if you give me some time for an interview so I can express my research idea in more detail.

Thanking you.

Sincerely,

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OBJECTIVE

Seeking for a challenging and rewarding research position within a dynamic group to utilize the skills developed through strong educational background and practical experience.

ACADEMIC QUALIFICATIONS

Degree	Institute	Year	Grade
Ph.D. Information and Communication Engineering	Harbin Engineering University, China	2016-2021	91 %
M.E. Telecommunication Engineering	Hamdard University, Karachi	2011-2013	3.2 (CGPA)
B.S. (Telecommunication Engineering)	FAST – National University, Karachi	2005-2009	3.1 (CGPA)
H.S.C.	D.H.A College	2005	First Division
S.S.C	Beacon House School System	2003	Grade 'A'

PROFESSIONAL EXPERIENCE

Ph.D. from Harbin Engineering University, China

September 2016 – December, 2021

Advisor: Prof. Qiao Gang, Prof. Liu Songzuo

Thesis Title: Covert Underwater Acoustic Communication Mimicking Humpback Whale Song

- Novel research on LPR constraint covert underwater acoustic communication mimicking sea natural sounds.
- **Published 08 SCI and 06 EI indexed journals (List attached).**
- **Invented 04 innovative techniques for covert communication mimicking Humpback whale song.**
- Gave course lectures of “**Modern communication**” to international students at Harbin Engineering University.
- Presented my research in various international conferences which includes **WUWNET 2018, ICTCA 2019, APPIIC 2020, IBCAST 2019 and 2020.**
- Co-supervised the research of Masters level students.
- **Reviewed research papers given by IEEE Access, International Journal of Communication System (IJCS), Hindawi Mobile Information Systems (MIS), HEARLD Aqua Fisheries and other journals.**
- Reviewed and proofread plenty of research papers, thesis and other technical articles of other students.
- Collaboration between Harbin Engineering University and Bahria University, NUST and Comsat University.
- Organized China Pakistan Marine Information (CPMI-2019) Seminar.

SENIOR MANAGER (TECHNICAL) at National Engineering and Scientific Commission

December 2016- present

- Leading a team of engineers to plan, execute and deliver marine research and technical projects.
- Design and development of covert underwater MODEM mimicking sea natural sound.
- Design and development of underwater vehicle.
- Indigenous design and development of HF and UHF MODEMs and its integration with legacy Radios.
- Research and development of technical projects including its feasibility, pre-sale and after sale support for Pakistan Navy and other marine organizations.
- Design and development of Software Defined Radio for HF and UHF range.
- Collaboration with academia locally and internationally for executing different defense and research projects.

ASSISTANT MANAGER (TECHICAL) at National Engineering and Scientific Commission

January 2011 – November 2016

- Design and developed under water imaging and illumination system with remote connectivity through Fiber Optic cable.
- Integrated different sensors (Gyro, compass, inclinometer, etc) with Moxa Serial multiplexer and remote operations through fiber optics cable for indigenously developed Unmanned Underwater Vehicle.
- Design, development, operations and upkeep of indigenously developed underwater cameras and lights.
- Replicate and fixed many faulty systems of Pakistan Navy which includes Flashing Light Buoy, Solar Panels, Smart Screens, etc.

ASSISTANT BUSINESS MANAGER at PTCL

October 2009 – January 2011

- Planning and optimization of Copper and Optical Fiber network (Installation of ONUs, Cabinet and DP) for future and current connections.
- Worked in network software CMS and BNCC to locate the complete path of a number, to resolve the high priority and pending complains.
- Resolving the issues and problems of copper and Optical Fiber Access Network including ONU/ Cabinet and ONT.
- Lead a team of Eight Foreman including their CSRs.

INTERNSHIP at WARID TELECOM IN ERICSSON CORE NETWORK

June 2008 – August 2008

- Worked in network software WinFIOL to perform call routing, locate location, and check subscriber data and services.
- Handled and solved Alarms and complains of BSC, BTS, MSC-S, MGW and HLR.
- Performed daily routine including making CDRs tapes and power routine.

RESEARCH PAPERS PUBLISHED

1. **Muhammad Bilal**, Songzuo Liu, Gang Qiao, Lei Wan, Yan Tao, “Bionic Morse Coding Mimicking Humpback Whale Song for Covert Underwater Communication” Applied Sciences 10 (1), 186, 2020. **Impact factor 2.679**
2. **Muhammad Bilal**, S Liu, G Qiao, W Raza, HH Zuberi. “Novel concept of bionic Morse coding for mimicry covert underwater communication” 2020 17th

- International Bhurban Conference on Applied Sciences and Technology (IBCAST).
3. G Qiao, **M Bilal**, S Liu, T Ma, Y Zhao, B Kong “Symmetry Oriented Covert Acoustic Communication by Mimicking Humpback Whale Song.” *Symmetry* 11 (6), 752. **Impact factor 2.713**
 4. G Qiao, **M Bilal**, S Liu, Z Babar, T Ma, “Biologically inspired covert underwater acoustic communication—A review” *Physical Communication* 30, 107-114, 2017. **Impact factor 1.81**
 5. G Qiao, **M Bilal**, S Liu, Q Liu, H Sun, “Covert Mimicry Communication using Humpback Whale Song” 2019 16th International Bhurban Conference on Applied Sciences and Technology (IBCAST).
 6. G Qiao, Y Zhao, S Liu, **M Bilal**, “Dolphin sounds-inspired covert underwater acoustic communication and micro-modem” *Sensors* 17 (11), 2447. **Impact factor 3.576**
 7. S Liu, M Wang, T Ma, G Qiao, **M Bilal**, “Covert underwater communication by camouflaging sea piling sounds” *Applied Acoustics* 142, 29-35. **Impact factor 2.639**
 8. Songzuo Liu, Habib Hussain Zuberi, Yi Lou, **Muhammad Bilal Farooq**, Shahabuddin Shaikh, Waleed Raza, “M-ary nonlinear sine chirp spread spectrum for underwater acoustic communication based on virtual time-reversal mirror method” *EURASIP Journal on Wireless Communications and Networking* 2021 (1), 1-20. **Impact factor 1.408**
 9. Gang Qiao, Tianlong Ma, Songzuo Liu, **Muhammad Bilal**, “A frequency hopping pattern inspired bionic underwater acoustic communication” *Physical Communication* vol. 46, 2021. **Impact factor 1.81**
 10. Xuefei Ma, Waleed Raza, Zhiqiang Wu, **Muhammad Bilal**, Ziqi Zhou, Amir Ali, “A Nonlinear Distortion Removal Based on Deep Neural Network for Underwater Acoustic OFDM Communication with the Mitigation of Peak to Average Power Ratio” *Applied Sciences* 10 (14), 4986. **Impact factor 2.679**
 11. W Raza, X Ma, T Wang, **M Bilal** “Low complexity energy efficient orthogonal frequency division multiplexing communication system over underwater acoustic channel by partial transmit sequence peak to average ratio.” *The Journal of the Acoustical Society of America* 146 (4), 2764-2765.
 12. G Qiao, Z Babar, L Ma, L Wan, X Qing, X Li, **M Bilal**, “Shallow water acoustic channel modeling and MIMO-OFDM simulations” 2018 15th International Bhurban Conference on Applied Sciences and Technology
 13. X Xuezhi, SM Ali, G Farid, **M Bilal**, “Image Processing In Visual Tracking By Various Techniques With The Use Of A Particle Filter– A Critical Review” *Journal of Flow Visualization and Image Processing* 23 (1-2).
 14. G Qiao, Q Liu, S Liu, **B Muhammad**, M Wen, “Symmetric Connectivity of underwater Acoustic Sensor Networks Based on Multi-Modal Directional Transducer”, *Sensors* 21 (19), 6548

15. ZS Liu, QJ Zhou, WS Gan, G Qiao, **M Bilal**, "Adaptive Joint Channel Estimation of Digital Self-Interference Cancellation in Co-time Co-frequency Full-Duplex Underwater Acoustic Communication", 2019 IEEE International Conference on Signal, Information and Data Processing (ICSIDP)

AWARDS AND ACHIEVEMENTS

- **Best international student of the year in Harbin Engineering University for the year 2017-2018.**
- Conducted a CPMI-2019 conference between China and Pakistan Experts at Harbin Engineering University, China.
- **Invited as guest speaker to deliver my novel research at C9 league Doctoral forum held in USTC, Hefei, China.**
- **Selected as the best social worker in Harbin Engineering University-International Students Union (HEU-ISU) of the year 2019-2020.**
- **Active Reviewer of journal articles at IEEE Access, Journal of Ocean Engineering, Sensors, Nature Scientific reports, International Journal of Communication System (IJCS), Hindawi Mobile Information Systems (MIS), HEARLD Aqua Fisheries and other journals**
- Worked voluntarily at HEU-ISU for four years and arranged tremendous extracurricular activities such as international cultural fusion, book shop, academic classes of MATLAB, ENDNOTE, AYN SYS, etc.
- Ran in Marathon for 21 km at Yichun, Heilongjiang, China.
- Won Second Position in **All ASIA NASCON 2008** in Engineering Problem Solving held at FAST-NU Islamabad.
- **Secured highest marks in TRW Technical Report Writing.**
- Resource manager of the university student community, DECS.
- Organized Hardware Competition in PROCOM' 07 and PROCOM.NET'08.
- Participated in SOFTEC' 08 and SOFTEC' 09 for microcontroller interfacing competition.
- Member of ACM, UKAN, IET and IEEE.

REFERENCES

1. Prof. Qiao Gang
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3. Prof. Zubair Shiekh
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